PROBLEMS SETS 2

1. In a survey of 60 people, 25 liked tea, 30 liked coffee, and 10 liked both. How many people liked only tea?

2. A survey of 100 students found that 70 students like pizza, 75 like burgers, and 60 like both. How many students like neither pizza nor burgers?

3. A class has 60 students. 35 students play football, 40 play cricket, and 15 play both. How many students play only one sport?

4. A survey discloses that in a group of 25 students, 20 are interested in music, 15 are interested in photography, 10 like sports, Further, 15 are interested in music and sports, 5 are interested in music and photography, 3 are interested in photography and sports and 2 are interested in all the three. Is the data correct?

5. In a school of 500 students, 65 do not play any game. 200 play cricket,180 play football, 260 play volleyball, 50 play both cricket and football, 85 play both football and volley ball, 100 play volleyball and cricket. Find
   (i) How many students play all three games?
   (ii) How many play only one game?

6. In a class consisting of 120 students, 30 take income tax, 40 take accounts and 45 take costing. 15 take income tax and accounts, 20 take income tax and costing, 12 take accounts and costing. 8 take all three subjects. How many do not take any of the three subjects? How many take only one subject?

7. A market research survey on the reading habits of 300 persons gave the following data in respect of these leading newspapers: 100 people read Times, 140 read Express, 125 read Hindustan, 50 read Times and Express, 30 read Express and Hindustan, 40 read Hindustan and Times and 20 read all of these. Find

   (i) The number of people who do not read any newspaper.
   (ii) The number of people who read just one newspaper.
   (iii) The number of people who read just two newspapers

8. Let X be the set of all three-digit integers such that $X = \{x \text{ is an integer}/100 \leq x \leq 999\}$. If A_i is the set of numbers in X whose i^{th} digit is i, Compute the cardinality of the set $A_1 \cup A_2 \cup A_3$.